Université Antonine
Faculty of Engineering and Technology
Department of Computer and Communication

aaq certified 2017-2022 Université Antonine
agence suisse d'accréditation et d'assurance qualité

Multimedia Database and Image Processing - Project Fall 2026 / 2027

An AI That Describes the World Out Loud

1. Description

Imagine walking into a building you have never seen, with no way to read the signs, find the door, or know who is standing in front of you. That is daily life for many blind and visually impaired people.

You will build a **voice assistant with eyes**. A phone camera streams live video. The user asks a question out loud, like "What's in front of me?" or "What did that sign say?", and hears the answer spoken back. Your system also **remembers what it recently saw**, so the user can ask, "What was written on the door I just passed?"

It is a project that mixes **AI**, **mobile**, **web** and **live streaming** into one demo.

> **Important:** This is a helper, not a replacement for a white cane or guide dog. Your assistant must never say that something is "safe", and it must admit when it is not sure.

2. Key Objectives

- Build a **multimedia database** that stores and searches images together with text, objects and time.
- Use **image processing** to pick good frames, improve poor ones and read text (OCR).
- **Stream live video** from a phone.
- Combine **speech, vision and generative AI** to give short, honest, spoken answers.
- Create an **accessible mobile app** and a **web dashboard**.

3. Content

Each team member leads one area, and everyone helps with integration.

Mobile App and Streaming

- Streams the phone camera live to your server (WebRTC, RTSP or HLS).
- Listens to the user's question and speaks the answer aloud.

Images and Multimedia Database

- Keeps only useful frames: sharp, new and informative.
- Reads signs and labels (OCR) and spots objects in each frame.
- Stores frames with text, objects and time, and finds them by time or by meaning.

Multimedia Database and Image Processing - Project Fall 2026 / 2027

AI Assistant

- Turns speech into text and finds the most relevant stored frames.
- A vision-language model writes a short answer based only on what the camera saw.
- Says "I'm not sure" when the evidence is weak, and works in at least two languages (English, French or Arabic).

Web Dashboard

- Shows live status, the timeline of stored frames, and past questions and answers.
- Displays Live Stream from device.

> **Ground rules:** No face recognition. Keep only what you need and delete old frames. Test only with people who agree.

4. Workflow

How the system works

Step	What happens
1. Stream	The phone sends live video to your server.
2. Remember	The server picks the useful frames, reads text, spots objects and saves everything in the database.
3. Ask	The user asks a question by voice.
4. Answer	Your system finds the right frames, the AI writes a short answer, and the phone speaks it aloud.

5. Deliverables

- **Code** in a Git repository with a README.
- **Short Presentation**: your design, technology stack, what you built and learned.
- **Test results:** Speed, search quality, answer accuracy and feedback.
- **Live demo** followed by questions for each member.
- **Team Responsibilities** what member developed what.

6. Bonus

- Warnings without being asked (closed door in front of you).
- Image pre-processing: Improving blurry or dark images before analysis.